

Applied Statistics (MATH10096)

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Introducing generalized linear models

↪ As we have seen extensively before, a linear model is a statistical model that can be written as

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

↪ Or, equivalently

$$E(y_i) \equiv \mu_i = \mathbf{x}_i^T \boldsymbol{\beta}, \quad y_i \stackrel{\text{indep.}}{\sim} N(\mu_i, \sigma^2).$$

Introducing generalized linear models

- ↪ **Generalized linear models** extend linear models by allowing some non-linearity in the model structure and much more flexibility in the specification of the distribution of the response variable y_i .
- ↪ A GLM has the basic structure:

$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}, \quad \mu_i \equiv \mathbb{E}(y_i),$$

where g is a smooth monotonic (i.e., differentiable and invertible).

- ↪ In addition, a GLM makes the distributional assumption that the y_i are independent and

$y_i \sim$ some exponential family distribution.

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- ↪ The **exponential family** of distributions includes many distributions that are useful for practical modelling: e.g., Poisson, binomial, gamma, and normal distributions.
- ↪ A convenient property of distributions in the exponential family, with the exception of the normal distribution, is that **the variance of y_i is a function of its mean μ_i** .
- ↪ A GLM therefore consists of three components:
 - ↪ The **systematic component**: $\mathbf{x}_i^T \boldsymbol{\beta}$.
 - ↪ The **random component**: the specified distribution for y_i .
 - ↪ The **link function** g .

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- ↪ Because generalized linear models are specified in terms of the *linear predictor* $\eta \equiv \mathbf{X}\beta$, many of the general ideas and concepts of linear modelling carry over with a little modification, to generalized linear modelling.
- ↪ Basic model formulation is much the same as for linear models, except that a link function and a distribution for the response variable must be chosen.
- ↪ Of course, if the identity function is chosen as the link function, along with the normal distribution for the response variable y , then the normal linear model we have covered before is recovered as a special case.

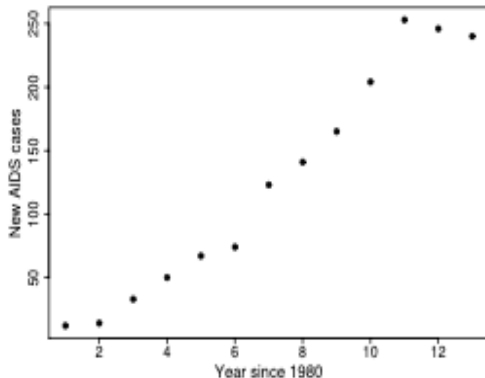
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- ↪ The generalization comes at some cost: model fitting now has to be done iteratively, and distributional results, used for inference, are now approximate and justified by large sample limiting results, rather than being exact.
- ↪ A brief historical note: generalized linear modelling as a methodology was developed in 1972 by John Nelder and Robert Wedderburn while working at the Rothamsted Experimental Station in the UK.

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AIDS in Belgium

→ New AIDS cases each year in Belgium, at the start of the epidemic.



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AIDS in Belgium

- ↪ In the early stages of an epidemic an exponential increase model is often appropriate.
- ↪ Hence, if μ_i is the expected number of new AIDS cases per year and if t_i denotes the number of years since 1980, then a suitable model might be

$$\mathbb{E}(y_i) \equiv \mu_i = \gamma e^{\alpha t_i}, \quad \gamma > 0$$

- ↪ Such a model can be turned into GLM form by using the log link so that

$$\log(\mu_i) = \log(\gamma) + \alpha t_i = \beta_1 + \beta_2 t_i, \quad \beta_1 = \log(\gamma), \quad \beta_2 = \alpha.$$

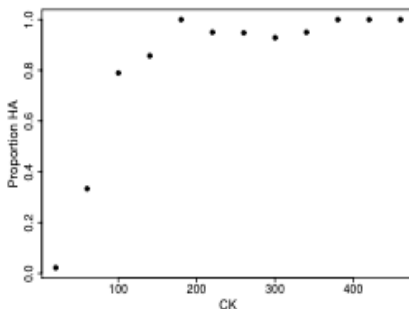
- ↪ The response variable is the number of new AIDS cases per year and, since this is a count, the Poisson distribution is probably a reasonable distribution to try.
- ↪ So the GLM for this situation uses a Poisson response distribution, log link function, and linear predictor $\beta_1 + \beta_2 t_i$

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Heart Attacks and Creatine Kinase

→ **Context:** study examining the efficacy of blood creatine kinase (CK) levels as a diagnostic when patients present with symptoms that may indicate a heart attack.

CK level	20	60	100	140	180	220	260	300	340	380	420	460
Heart Attack	2	13	30	30	21	19	18	13	19	15	7	8
Not Heart Attack	88	26	8	5	0	1	1	1	1	0	0	0



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Heart Attacks and Creatine Kinase

- ↪ Let p_i be the probability of heart attack at CK level x_i .
- ↪ A particularly convenient model for describing this probability is

$$p_{x_i} \equiv p_i = \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}.$$

- ↪ This curve is sigmoid in shape, and bounded between 0 and 1.
- ↪ If y_i is the number of heart attack victims observed out of N_i patients with CK level x_i , we have that $y_i \sim \text{Binomial}(N_i, p_i)$ and therefore,

$$\mathbb{E}(y_i) \equiv \mu_i = N_i p_i = N_i \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}.$$

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Heart Attacks and Creatine Kinase

- This model is somewhat non-linear in its parameters, but if we apply the **logit** link function we obtain that

$$\begin{aligned}g(\mu_i) &= \log \left(\frac{\mu_i}{N_i - \mu_i} \right) \\&= \log \left(\frac{N_i p_i}{N_i - N_i p_i} \right) \\&= \log \left(\frac{p_i}{1 - p_i} \right) \\&= \log \left(\frac{\frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}}{\frac{1}{1 + e^{\beta_1 + \beta_2 x_i}}} \right) \\&= \log \left(e^{\beta_1 + \beta_2 x_i} \right) \\&= \beta_1 + \beta_2 x_i\end{aligned}$$

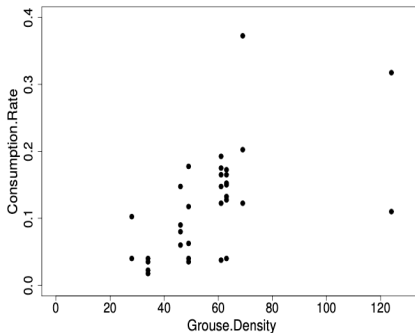
which is clearly a GLM.

- This model is known as **logistic regression**.

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Hen harriers and Grouse

↪ Daily consumption rate of Grouse by Hen Harriers (a type of bird of prey) plotted against the density of Grouse on various Grouse moors.



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Hen harriers and Grouse

- ↪ If c_i denotes consumption rate and d_i is grouse density, then ecological theory suggests that the expected consumption rate should be related to grouse density by the following model

$$\mathbb{E}(c_i) \equiv \mu_i = \frac{\alpha d_i^3}{\delta + d_i^3}.$$

- ↪ Here α and δ are parameters to be estimated.
- ↪ Obviously this model is nonlinear in its parameters.
- ↪ However by using the 'inverse' link function, the right hand side of the previous equation can be made linear in the parameters

$$\frac{1}{\mu_i} = \frac{1}{\alpha} + \frac{\delta}{\alpha} \frac{1}{d_i^3} = \beta_1 + \beta_2 \frac{1}{d_i^3}, \quad \beta_1 = \frac{1}{\alpha}, \quad \beta_2 = \frac{\delta}{\alpha}.$$

- ↪ Because the variability in Grouse consumption rate likely increases with the mean rate, suggesting a gamma distribution for the response, and completing the model specification.